

Service Specifications

AI Service (Heterogenous System Test)

Isolated service for direct output requests and inter-service data passing.

- **Constraint:** Requires maximum load caps per request and strict isolation due to extreme resource intensity.
- **Use Case:** May be utilized for test data generation post-normalization.
- **Dependencies:**
 - **Filter Service:** If generating identical pictures.
 - **Permutation Service:** Computes $n!$ complexity without affecting execution time dependencies.

Media Services

- **Video Service:** Handles media compression and frame-by-frame splitting. Communicates downstream with the **Digital Filters Service** to apply digital visual filters.
- **Digital Filters Service:** Executes complex matrix manipulations. Can be separated into specific standalone filters if needed.

Mathematical Modules

Abstract modules used for controlled load testing and stressing system communications:

- **Differential Equations Service:** Solves differential equations with varying complexity to test inter-service communication and load handling.
- **Integration Service:** Heavy computing of integrals.
- **Permutations Service:** Finds all permutations of a given set ($O(n!)$). Designed to severely load a single node.
- **Fibonacci Service:** Used specifically to investigate CPU clock scaling based on request complexity.

System Observability & Metrics

Performance Targets

- **Throughput:** Media files routing
- **CPU Clock:** Complex math calculations (Math Node 1)
- **RAM Usage:** State/Cache heavy math (Math Node 2)
- **Latency:** Monitored globally across all services

Infrastructure Focus

- **Observability:** Analyzing the performance delta between native implementations and Service Mesh architectures.

